



Alternating Optimization for LLM Post-Training: From Divergence to Verified Solutions

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Research Objectives

- Diagnose **why** ALS-based post-training diverges on deep LLMs (≥ 28 layers) while converging on shallow models.
- Test three natural repairs — gradient injection, low-rank probing, soft-target distillation — with matched-budget controls.
- Identify **verified alternatives** within a 2×2 factorial framework (optimizer $\times$ parameter form).

Background & Problem

Alternating Least Squares (ALS) replaces hundreds of gradient steps with one closed-form solve $W^* = (X^T X + \lambda I)^{-1} X^T Y$ on the output layer. Applied to LLM post-training, ALS works on ≤ 24 -layer models but **diverges on every 28-layer attempt** (11/11 on Qwen2.5-7B, PPL $\rightarrow$ NaN within 1–3 cycles).

A controlled five-condition ablation on Qwen2.5-7B isolates the cause: **only conditions containing real ALS weight modification diverge**; SGD-only, perturbation-only, and ALS-with-suppressed-weights all converge on the same model (Fig. 5).

Methods

2×2 factorial framework. ASP vs AdamW, crossed with full-rank vs LoRA (protocols A–D), compared at equal FLOPs budgets with shared data, seeds, and evaluation.

Causal analysis. Residual connections amplify an ALS perturbation by $\rho \approx 1.08$ per layer ($1.08^{27} \approx 8 \times$ at 28 layers) — far beyond SGD's per-cycle recovery (≈ 0.005), predicting the depth boundary $L_{\max} \approx 26$.

Harness audit. OPT-scale numbers re-verified with the correct tokenizer and pad-masked labels (pre-trained baseline corrected to 73.7 PPL).

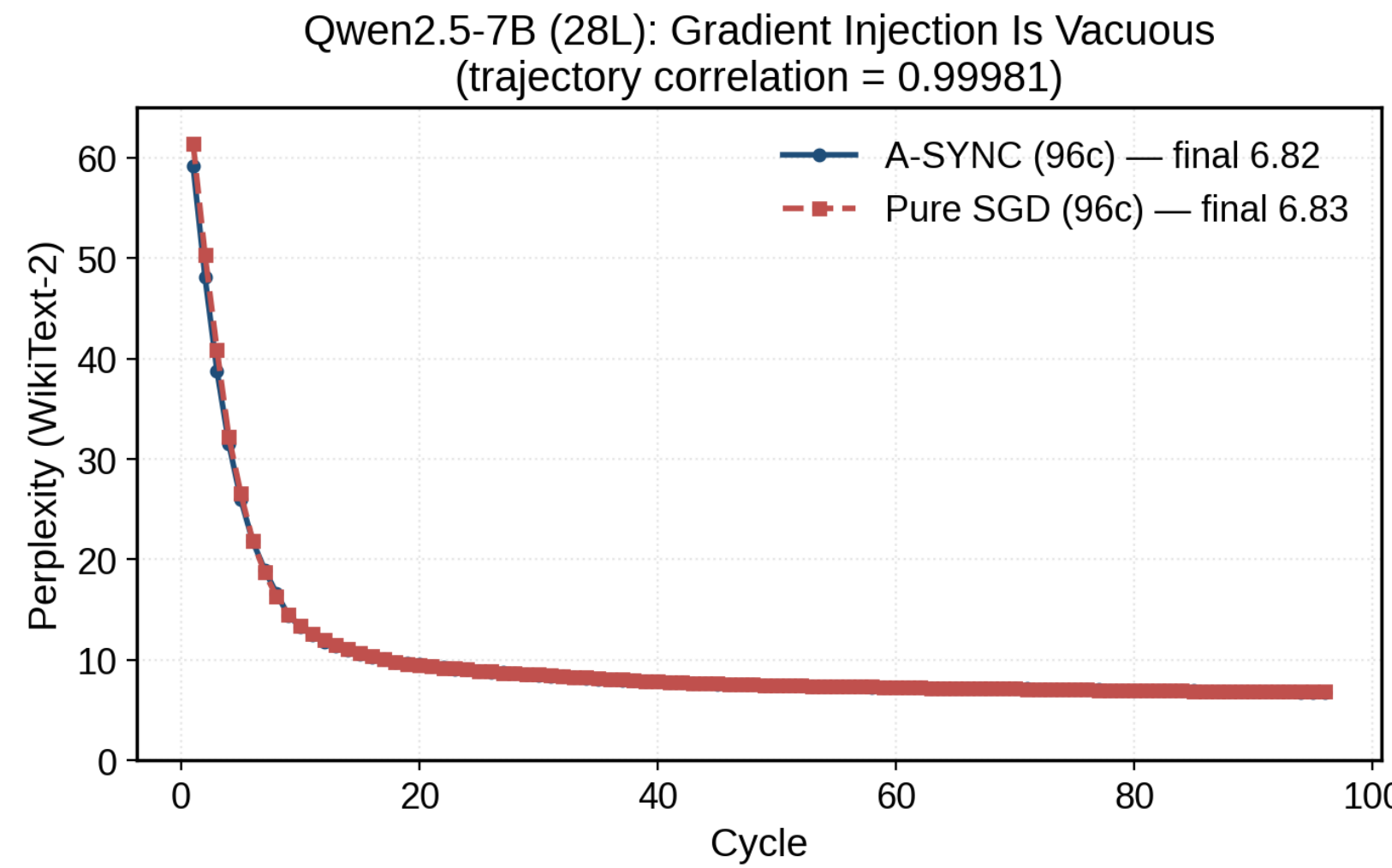
Highlights

- First proof (not assumption) that ALS weight modification — not noise, not optimizer mismatch — is the sole sufficient cause of divergence.
- A falsifiable residual-amplification theory with predictive depth boundary.
- Rigorous negative result: three repair designs all fail matched-budget controls.

Key Findings & Experimental Results

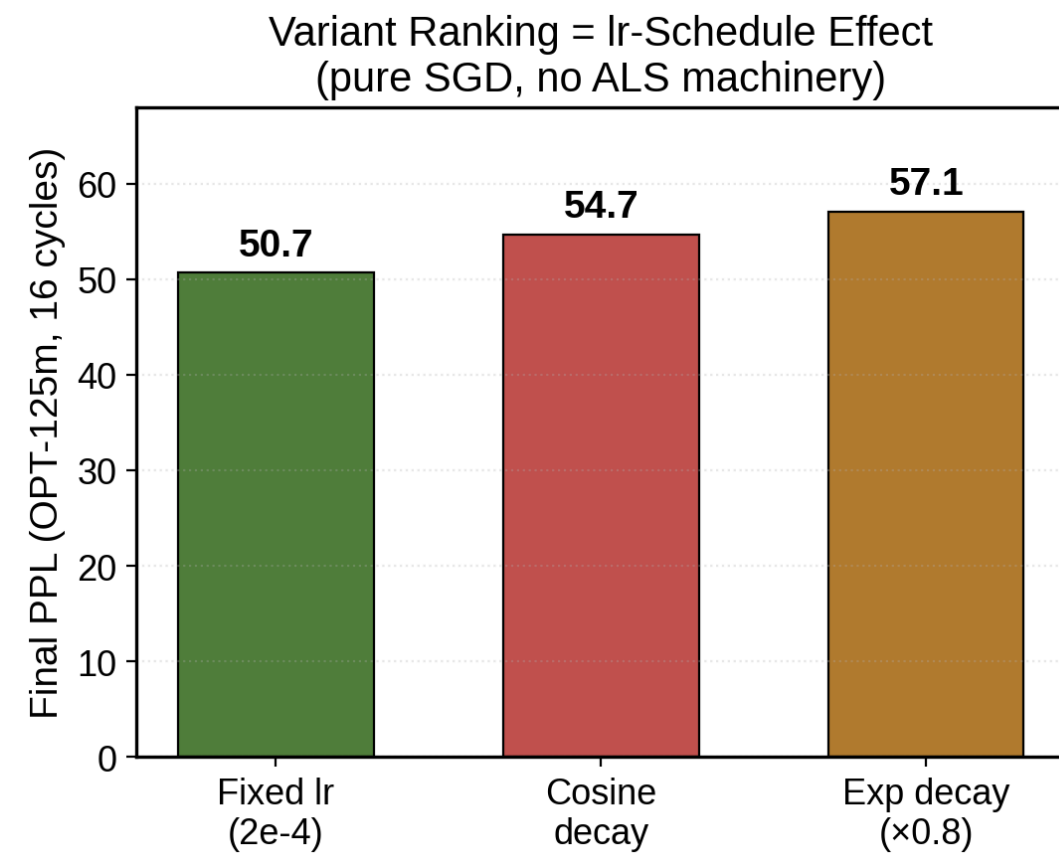
Finding 1 — Gradient Injection Is a Timing No-Op

A-SYNC injects the ALS solution as a gradient bias; matched-budget pure SGD reproduces its trajectory almost exactly (correlation 0.99981). The injection never beats plain SGD.



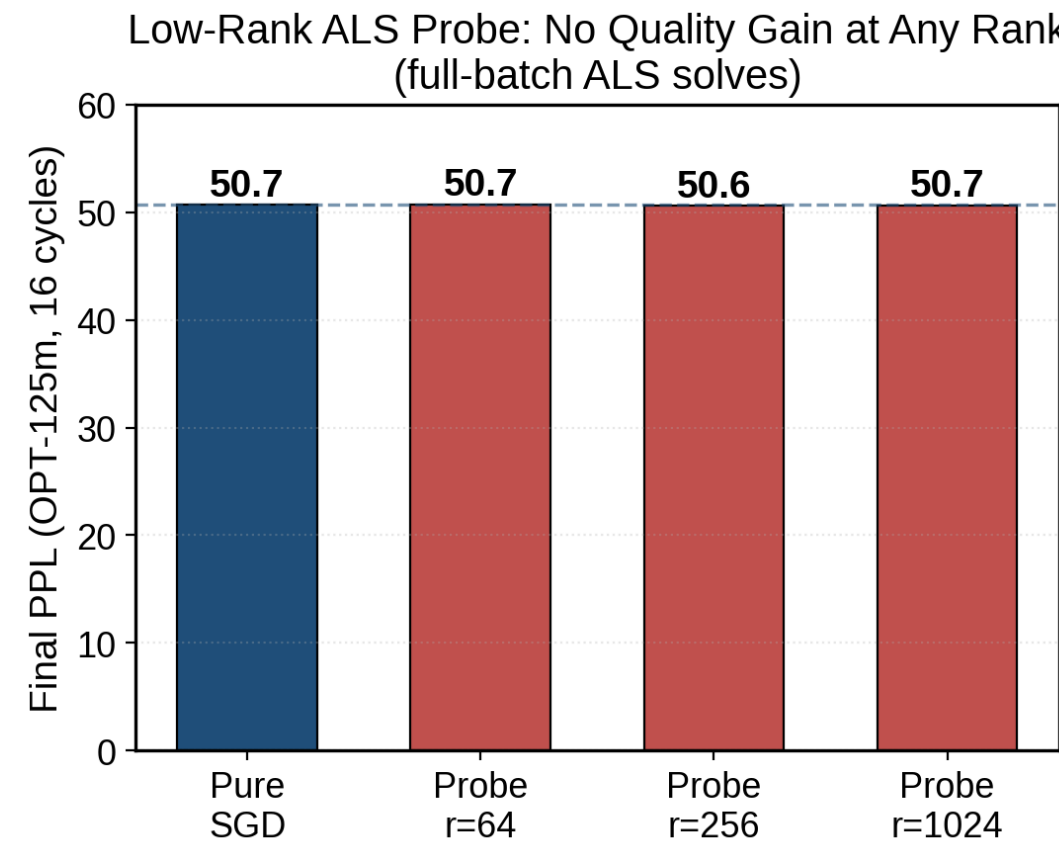
Finding 2 — The Variant Ranking Is an lr-Schedule Effect

With injection proven vacuous, the perceived algorithm ranking is explained by the learning-rate schedule alone (pure SGD, no ALS machinery).



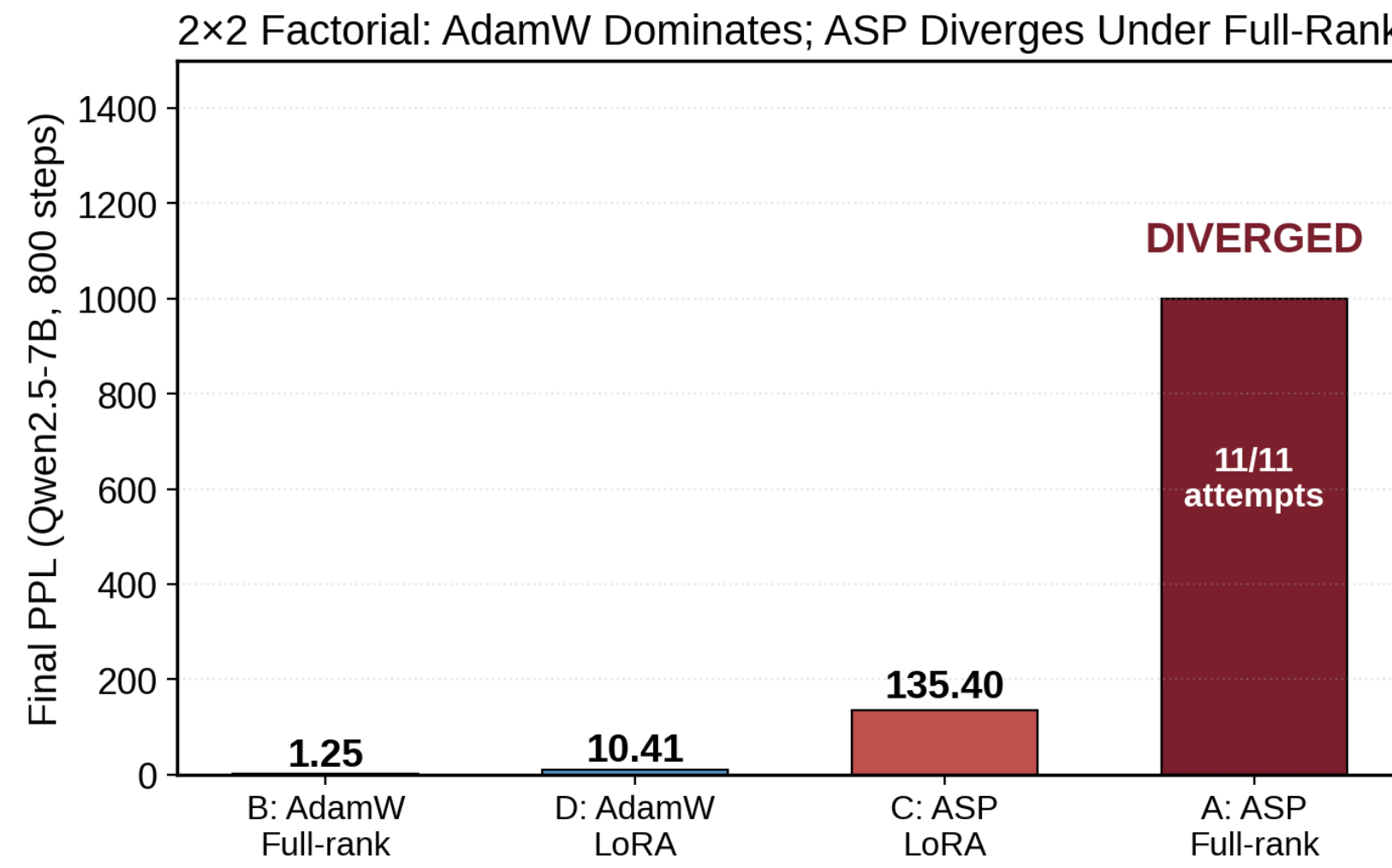
Finding 3 — Low-Rank ALS Probes Add No Quality at Any Rank

Widening the probe bottleneck ($r = 64/256/1024$, full-batch ALS) never beats pure SGD — the closed-form solve is redundant with the CE-gradient direction.



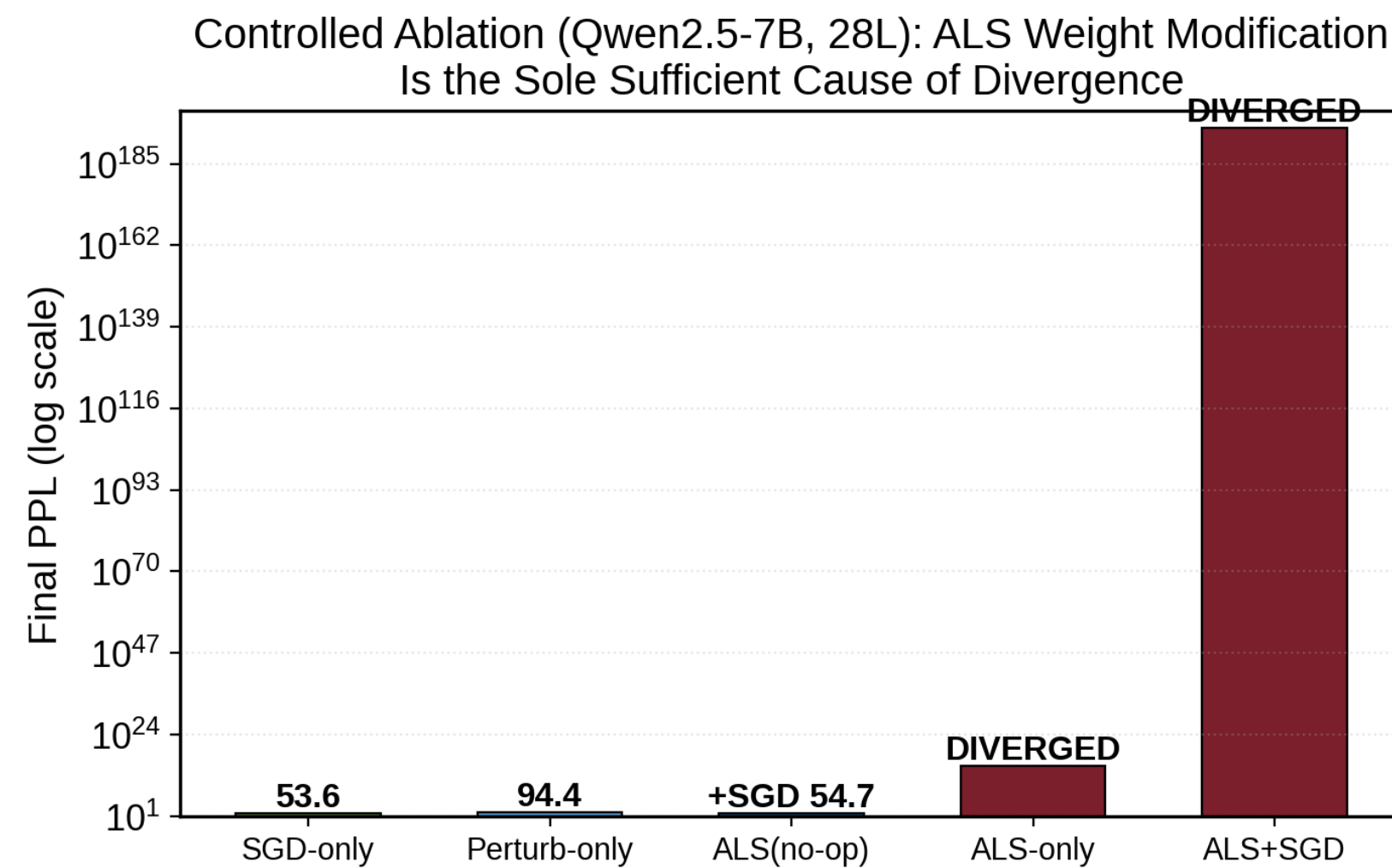
Finding 4 — The 2×2 Matrix: AdamW Dominates, ASP Diverges

On Qwen2.5-7B (800 steps, $N=3$): AdamW full-rank 1.25 PPL, AdamW+LoRA 10.41, ASP+LoRA 135.4, ASP full-rank diverges.



Finding 5 — ALS Weight Modification Is the Sole Cause of Divergence

ALS-only diverges in a single solve ($73 \rightarrow 2.0 \times 10^9$ PPL); every ALS-free condition converges. Soft-target (distillation) ALS adds no value either (52.6 vs 52.4 SGD).



Verified Solutions (Qwen2.5-7B, 28 layers)

Plain SGD, fixed lr — 6.83 PPL in 4800 steps (power-law tail, asymptote ≈ 3.5). The ALS-free baseline that converges where ALS cannot.

AdamW full-rank — 1.25 PPL ($N=3$). Best absolute quality.

AdamW + LoRA — 10.41 PPL with $\sim 0.1\%$ trainable parameters. Best compute efficiency.

Practical takeaway: on deep models, remove the ALS machinery and let gradient descent do its job with a sustained learning rate.

References

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